Geotemporal Graph Modeling for Effective Coverage in Multi-hop Vehicular Networks

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Abstract

Achieving reliable message dissemination in multi-hop vehicular networks (VANETs) is critical for safety, disaster management, and decentralized coordination. However, the highly dynamic topology of these networks often disrupts end-to-end connectivity, making adequate coverage a significant challenge. While traditional routing strategies focus on node-centric metrics, this paper argues that the network's underlying spatial structure is a more stable predictor of connectivity. We propose a novel dissemination strategy that partitions the network into spatial regions (tiles) and identifies improved source zones to maximize message reachability. We model the network as a geotemporal graph—a representation that explicitly captures spatiotemporal dependencies in propagation dynamics. Using both real-world (πNEUMA) and synthetic (SUMO) vehicular traces, we demonstrate that a small, identifiable subset of regions consistently acts as superior injection points for message broadcasting. Our heuristic algorithm leverages these geotemporal properties to significantly improve network coverage compared to traditional methods, offering a scalable solution for infrastructure placement and data dissemination protocols.

Keywords:
Multi-hop vehicular networks, reachability, geotemporal graphs, source regions, heuristic algorithms.

1. Introduction

Ensuring high reachability is a critical requirement for data dissemination in multi-hop vehicular networks, mainly when the information must spread over the largest possible area, and latency plays a significant role [1, 2]. The consequences of instability in message reachability are visible in public security scenarios. For example, poor deployment of vehicular communication infrastructures – such as fixed relay stations, mobile towers, or drone-based broadcasters – may result in urban areas without coverage or excessive latency, threatening public safety, evacuations, and first responder coordination. As mobile networks shift toward decentralized architectures, strategic data dissemination becomes crucial for improving reachability [3, 4, 5]. However, achieving effective reachability remains challenging due to the inherently dynamic and unstable topologies of vehicular networks, which prevent the establishment of guaranteed end-to-end paths and complicate message dissemination [6, 7, 8].

Traditional routing and placement methods address this problem by evaluating a node's importance based on network characteristics, such as its topology [9]. However, in dynamic networks, mobility is strongly influenced by the physical space occupied by nodes. For example, vehicular networks are confined to road layouts, while pedestrian networks mainly occupy sidewalks. These city structures, with recurring traffic patterns and human behavior cycles, shape mobility [10, 11]. Regarding message spreading, prior work has predominantly focused on node-centric routing, identifying influential nodes based on their metrics and designing protocols based on opportunistic encounters [12, 13]. While some protocols incorporate spatial factors (e.g., SPread and ASpread, which leverage social communities and geographic locality in vehicular delay-tolerant networks to improve routing efficiency), these approaches still prioritize node importance over explicit spatial analysis [14]. That said, the performance gains from adding geographic considerations in ASpread suggest that spatial regions themselves may play a fundamental role in message dissemination.

We shift the focus from individual nodes to geographic regions as the primary unit of network analysis. We propose a method that partitions the target area into cells (or tiles) and selects an improved subset of them to serve as message injection points. This region-centric approach explicitly leverages the spatial structure of vehicular networks, offering a complementary strategy to traditional store-carry-forward methods. More specifically, the central problem we tackle is the following: *given a large-scale vehicular trace and a constraint on communication range and delay, how can one systematically select a set of n tiles, which will act as sources, with improved message spatial reachability?* In this context, source tiles refer to specific places, well-suited for deploying source nodes that may disseminate a message. Figure 1 illustrates an example of a multi-hop vehicular network scenario with three different source regions, highlighting their connection ranges, reachability, and also where they complement or intersect each other.

We introduce three key contributions. First, we propose a

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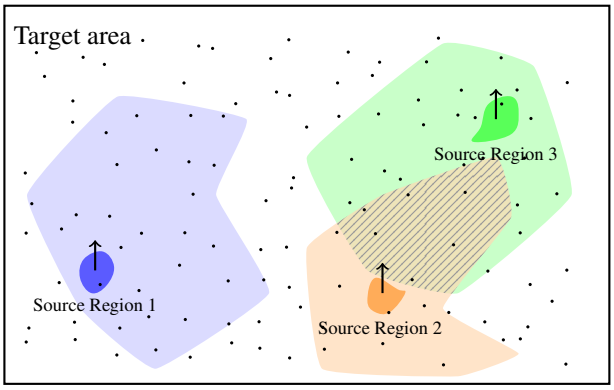


Figure 1: Example of spatial reachability of multi-hop messages starting from 3 different regions, indicated as arrows. The colored areas represent the space that each region can reach, and the hatched areas belong to two different regions concurrently. Each dot represents a vehicle that can be *reached* if it is inside a colored area.

representation of multi-hop vehicular networks as *geotemporal graphs*, which explicitly integrates both geographical constraints and temporal dynamics into a unified network model. Unlike traditional temporal graph models that treat space implicitly, our representation explicitly captures how physical proximity and spatial structure influence link availability over time, enabling more accurate modeling of propagation dynamics in geographically constrained environments. Second, we develop a quantitative evaluation method to assess the dissemination potential of the proposed strategy. The corresponding metric characterizes how effectively messages originating from a given region propagate throughout the network, accounting for the spatial extent of coverage and the temporal evolution of the network topology. By quantifying region-centric reachability rather than node-centric influence, our approach reveals spatial patterns that would otherwise remain hidden in node-centric analyses. Third, we propose a heuristic algorithm specifically designed to identify and select improved source regions that maximize message spatial reachability. Our method systematically evaluates candidate regions within the geotemporal graph framework and selects a strategic subset that collectively ensures broad and efficient message coverage. As a consequence, it offers a practical approach for infrastructure deployment (e.g., RSU placement) and target selection in dissemination protocols.

We validate our approach using two complementary vehicular datasets: one synthetic, generated with SUMO [15], and a real-world dataset known as πNEUMA [16]. Both datasets are transformed into geotemporal graphs using our proposed framework, enabling a unified analysis of spatiotemporal dependencies and demonstrating the generalizability of our methodology across diverse network conditions. Our experimental results reveal two significant findings. First, we identify that only a small, distinct subset of spatial regions consistently exhibits superior message dissemination capabilities across different time periods and network conditions. Observations suggest that spatial structure plays a predictable and exploitable role in the propagation of messages. Second, we demonstrate that our heuristic algorithm successfully selects message sources with enhanced reachability and ensures the scalability of the set size. These results confirm that strategically selecting source regions based on geotemporal analysis offers tangible benefits for practical deployment in vehicular applications.

The remainder of this paper is as follows. Section 2 formally defines the problem of message reachability in vehicular networks, identifies key research gaps, and introduces our geotemporal reachability framework. Section 3 presents our methodology for constructing geotemporal graphs from vehicular traces. Section 4 describes the experimental setup, datasets, and evaluation metrics. We then present our empirical analysis in two stages. Section 5 examines geotemporal reachability patterns across spatial regions using both synthetic and real-world vehicular datasets, revealing which regions consistently exhibit superior propagation capabilities. Section 6 introduces and evaluates our heuristic algorithm for selecting improved source region sets. Section 7 situates our work within the broader literature. Finally, Section 8 summarizes our contributions and outlines promising directions for future research.

2. Reachability in Geotemporal Graphs

We aim at representing a vehicular network in a way that supports analyses of both temporal and spatial correlations. Instead of modeling the network through nodes and links between them, we abstract the mobility patterns and interactions over time and space through a *tiling-based temporal graph*, which we refer to as a *geotemporal graph*. This abstraction enables not only the adaptation of dense mobility traces in a scalable fashion but also supports reasoning over regions rather than individual nodes. We proceed by analyzing its temporal reachability, refining its definitions, and close the section with a heuristic algorithm that selects tile sources to enhance reachability.

2.1. Geotemporal Graphs

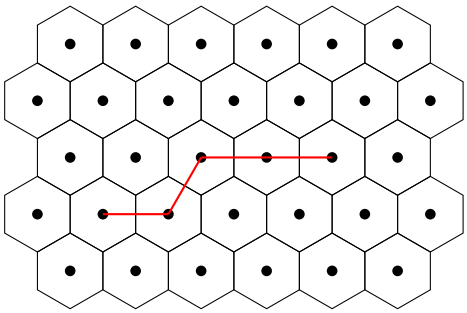
To explore the spatial characteristics, one crucial step is to divide the space into manageable, regular units. We adopt tiling as a tool to enable this abstraction.

Let $\mathcal{T} = \{T_1, T_2, \dots\}$ be a regular tiling of the Euclidean plane, where each tile T_i is a convex polygon that shares edges with a finite number of neighboring tiles and has a center point $c_i \in \mathbb{R}^2$. Let $\mathcal{R} \subset \mathbb{R}^2$ denote a bounded region of the plane. We define the *induced tiling of target region* $\mathcal{R}$, denoted by $\mathcal{T}_\mathcal{R} \subset \mathcal{T}$, as the subset of tiles such that:

1. $T_i \in \mathcal{T}_\mathcal{R} \Rightarrow T_i \cap \mathcal{R} \neq \emptyset$, i.e., each tile intersects the target region $\mathcal{R}$.

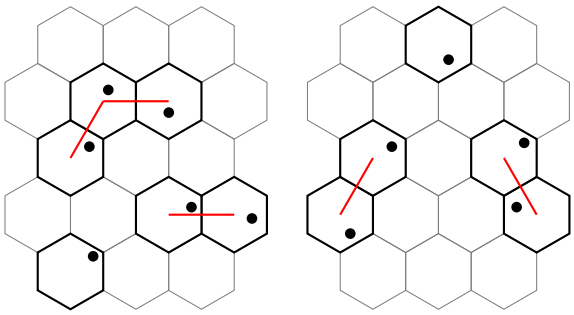
2. $\mathcal{T}_\mathcal{R}$ is minimal under inclusion, i.e., it contains only those tiles in $\mathcal{T}$ that have a non-empty intersection with $\mathcal{R}$.

Once we have organized the space into tiles, we must define how to measure proximity and distance between them. We adopt a notion of distance based on tile adjacency. We define



$t = 0$

Figure 2: Regular tiling of a region with hexagonal polygons and a 4-hop distance between two tiles.



$t = 0$ $t = 1$

Figure 3: Example of a geotemporal graph with $\theta = 1$ as two static graphs, one at $t = 0$ and another at $t = 1$. The thin edges represent the region-induced tiling and bold edges the occupied tiles.

two tiles $T_i, T_j \in \mathcal{T}$ to be *adjacent* if they share a common edge. Given this adjacency relation, the *hop distance* between two tiles T_i and T_j, denoted $d_{\mathcal{T}}(T_i, T_j)$, is the minimum number of adjacent-tile transitions required to reach T_j starting from T_i. For example, Figure 2 represents a 4-hop distance between cells i and j. If there is no such finite sequence of adjacent tiles connecting T_i and T_j, we define $d_{\mathcal{T}}(T_i, T_j) = \infty$.

Since the activity of the vehicles defines the network and it only exists in their presence, it is necessary to determine which tiles are "active" at any given moment. These are the tiles that at least one vehicule occupies at a given time step. Let $\mathcal{M} = \{m_1, m_2, \ldots, m_k\}$ be a set of vehicles, where each node m_i has a known position $p_i(t) \in \mathbb{R}^2$ at each discrete time step $t \in \mathcal{T}_{\text{time}}$. At time step t, the *occupied tile set* of the target region $\mathcal{R}$ is:

$$O_t = \{T_i \in \mathcal{T}_{\mathcal{R}} \mid \exists m_j \in \mathcal{M},\ p_j(t) \in T_i\}. \tag{1}$$

Equation 1 defines a dynamic spatial footprint of the network over time, based on the distribution of nodes.

With space tiling, tile adjacency, and occupancy established, it is possible to define a geotemporal graph representation that abstracts node-to-node interactions. Instead, this graph captures the evolution of connectivity between spatial regions (here represented by tiles) based on the presence and movement of vehicles. By focusing on which regions are active and how close they are to one another, it becomes possible to model the structural dynamics of the network in a way that integrates both spatial and temporal dimensions. This geotemporal abstraction is particularly well-suited for understanding the collective behavior of mobile systems, such as message reachability, while it facilitates scalability and readability. It also enables the study of how mobility patterns and spatial configurations influence the emergence of connectivity over time, providing a foundation for the analyses presented in this paper.

Definition 1 (Geotemporal graph)**.** The *geotemporal graph* at time t is defined as:

$$\mathcal{G} = (\mathcal{G}(t))_{t \in \mathcal{T}_{time}} = (V_t, E_t), \tag{2}$$

where the time domain is the same of $\mathcal{M}$ ($t \in \mathcal{T}_{\text{time}}$), the vertices are the occupied tiles at each time step ($V_t = O_t$), and the edge $(T_i, T_j) \in E_t$, if and only if $T_i, T_j \in V_t$ and $d_{\mathcal{T}}(T_i, T_j) \leqslant \theta$, where $\theta \in \mathbb{N}$ is a fixed threshold parameter that represents the communication range in number of hops. This graph corresponds to a sequence of static graphs $(G_t, G_2, ..., G_{\mathcal{T}_{\text{time}}})$, where $G_t = (V_t, E_t)$. For instance, Figure 3 represents a geotemporal graph as one static graph for each time step.

2.2. Time-Aggregated Reachability Graph

In geotemporal graphs, message propagation must respect spatial and temporal constraints. One message cannot travel instantaneously, and more importantly, it must respect temporal causality.

A time-respecting path is a sequence of couples $\mathcal{J} = \{(e_1, t_1), (e_2, t_2), \ldots, (e_k, t_k)\}$ such that, $\forall i < k$:

1. e_i arrives in the same vertex that e_{i+1} departs.
2. e_i exists at time t_i; formally, $e_i \in E_{t_i}$.
3. Messages do not travel instantaneously, i.e., $0 < t_{i+1} - t_i$.

Also, $\mathsf{departure}(\mathcal{J})$ and $\mathsf{arrival}(\mathcal{J})$ denote the starting time t_1 and the finishing time t_k, respectively.

We now introduce the concept of a *time-aggregated reachability graph*. This graph captures which vertices can be reached from others under a time constraint δ, using time-respecting paths as defined above. For $\delta \in \mathbb{R}+$, we define the reachability graph with maximum delay δ as $\mathcal{R}_\delta = (\mathcal{R}_\delta(t))_{t \in \mathcal{T}_{time}}$ as follows. For all t, the edge $(u, v) \in \mathcal{R}_\delta(t)$ if and only if there exists at time t in $\mathcal{G}$ a journey $\mathcal{J}$ from u to v such that $\mathsf{departure}(\mathcal{J}) \geqslant t$ and $\mathsf{arrival}(\mathcal{J}) \leqslant t + \delta$. In the geotemporal context, this graph encodes all valid message journeys, between different regions, that begin at time t and end within time $t + \delta$.

Building on the previous definition, one may need to account for vertex reachability across the entire temporal domain. The time-aggregated reachability graph summarizes, over time, how often a region is reachable from another. It is defined as the complete weighted graph $\mathcal{TR}_\delta = (V, E, c)$, where:

1. V is the set of all distinct tiles that appear in the geotemporal graph over the entire time domain:

$$V = \bigcup_{t \in \mathcal{T}_{\text{time}}} V_t.$$

2. $E = V \times V$ is the set of all ordered pairs of tiles.

3. $c : E \to [0,1]$ is the *time-aggregated reachability function*, which maps each edge $(u,v) \in E$ to the fraction of time steps in which tile v can be reached from tile u in the corresponding reachability graph. Formally:

$$c(u,v) = \frac{1}{|\mathcal{T}_{\text{time}}|} \sum_{t \in \mathcal{T}_{\text{time}}} \mathbf{1}_{(u,v) \in \mathcal{R}_\delta(t)},$$

where $\mathbf{1}_{(u,v)\in\mathcal{R}_\delta(t)}$ is the indicator function equal to 1 if v is reachable from u at time t, and 0 otherwise.

When we represent the time-aggregated reachability graph as an adjacency matrix, each row represents a candidate source tile, and each column represents another tile in the given spatial region. The entry at row i, column j indicates the percentage of time that tile i covers tile j throughout the temporal window. Consequently, each row, referred to here as the time-aggregated reachability vector, represents the extent to which the respective tile reaches the target region over time.

To evaluate the performance of a tile or a set of tiles, we compare its time-aggregated reachability vector with the best-case scenario by measuring the difference between them. We expect the best set of tiles to consistently reach the entire map over time, represented here as a vector of ones for each tile in the region.

When considering the time-aggregated reachability vector of a set of tiles, we calculate a lower bound estimate for their combined time-aggregated reachability by taking the element-wise maximum between their vectors. To this end, we leverage the following property: if tile u covers tile w for c_1 percent of the time, and tile v covers w for c_2 percent of the time, then w is covered by at least one of them for $\max(c_1, c_2)$ percent of the time, which then gives a lower bound for reachability. Formally, the time-aggregated reachability vector of a tile set $\mathbb{S}$ is defined as:

$$\mathcal{V}_{\mathbb{S}}[T_j] = \max_{T_i \in \mathbb{S}} c(T_i, T_j) \mid T_i, T_j \in V(\mathcal{TR}_\delta). \tag{3}$$

Then, we measure the difference between the two vectors through a mean squared error. Let $\mathcal{V}_{\mathbb{S}}$ be the time-aggregated reachability vector of a tile set $\mathbb{S}$, and $V = V(\mathcal{TR}_\delta)$. Finally, we define a metric called the *time-aggregated reachability error*, which condenses the spatiotemporal performance of a set of tiles for the entire spatiotemporal domain:

$$\mathcal{E}(\mathcal{TR}_\delta, \mathbb{S}) = \frac{1}{|V|} \sum_{T_i \in V} (1 - \mathcal{V}_{\mathbb{S}}[T_i])^2. \tag{4}$$

2.3. Heuristic Algorithm for Source Set Construction

The task of selecting the best set of n source tiles to improve spatial reachability is a combinatorial problem. Given a universe of candidate tiles, the number of possible source sets of size n grows exponentially, specifically as $\binom{|V|}{n}$. As a result, computing the optimal combination through brute-force search becomes unmanageable for realistic scenarios.

To tackle this challenge, we propose an algorithm that builds the source set incrementally. At each step, the algorithm selects the tile that offers the highest gain in reachability, measured by a heuristic score, when added to the current set. At this point, we represent the time-aggregated reachability graph as an adjacency matrix, as discussed in Subsection 2.2 above.

Starting with an empty source set and its time-aggregated reachability vector, the algorithm follows these steps:

1. For each candidate tile not yet in the source set, the algorithm calculates the time-aggregated reachability of the union with the current source set, as described in Subsection 2.2.

2. Among all candidate tiles, the algorithm calculates all possible unions with the current source set, and selects the one that minimizes the time-aggregated reachability error.

3. The selected tile is then added to the source set, and the current time-aggregated reachability vector is updated accordingly. This process repeats until n tiles have been selected.

3. Building a Geotemporal Graph

We now describe our procedure for constructing geotemporal graphs from vehicular traces. Building on the previous formal concepts, we transform raw vehicular data into a sequence of static graphs representing the dynamic connectivity between regions. This method adapts to different region sizes and connection distances, and applies to any mobility dataset where node positions change over time.

The input is a vehicular trace, a list of events that record each vehicle's position at specific times. Each entry is a tuple *(id, lat, lon, time)*, where *id* uniquely identifies a vehicle and *(lat, lon)* specifies its position at time instant *time*. These observations may originate from GPS devices, cellular data, or video tracking. Since collection devices often record at irregular intervals, measurements must be discretized into regular time windows to enable consistent graph construction at each time step.

To enable consistent graph construction, we discretize time into uniform duration windows. All observations within each window are aggregated to form the graph snapshot. The choice of the window determines the temporal granularity of the model. Smaller windows capture finer-grained dynamics but increase computational cost for temporal path calculations, while larger windows smooth over rapid topology changes, potentially missing transient connectivity opportunities.

Concerning spatial discretization, we use the H3 geospatial indexing system developed by Uber [17]. The H3 system offers a hierarchical and hexagon-based tiling of the Earth's surface, well-suited for modeling vehicular data and connectivity. Unlike square grids, which can suffer from directional bias or uneven distances between neighbors, hexagonal grids offer a more balanced space division and are especially suited for urban environments. Each tile is encoded in H3 by a unique hierarchical geospatial index corresponding to a position and resolution.

The resolution determines the mean edge length of hexagonal tiles, allowing users to trade off between spatial granularity and computational cost.

We rely on the official H3 library to map each coordinate to its tile index. This mapping results in a set of tiles characterized by the time that they are active (the occupied tile set). This temporal sequence of tile sets describes the evolution of a region's vehicular footprint. Abstracting node-to-node interactions reduces computational costs and provides an interpretable representation of area occupancy within the network at a particular spatial granularity.

We can now express connectivity between tiles. A connection distance parameter, expressed in meters, represents the maximum distance over which two tiles may communicate. In a real setup, it would correspond to the communication range of the vehicles. This distance maps into a compatible metric in H3, represented by θ, as an upper limit in the hop distance between cells. Subject to a resolution, this constant is calculated by taking the integer part of the division between the chosen connection distance and the average center-to-center distance between adjacent tiles.

Once the occupied tile sets and the connection range θ are known, we can construct the geotemporal graph. For each time step t, a graph is built by taking the occupied tile set at t as vertices and introducing edges between all tiles with a hop distance less than or equal to θ. This produces a stream or sequence of static graphs, each indexed at a specific t, that together model the temporal evolution of spatial relations induced by the vehicles. Structuring vehicular traces as geotemporal graphs provides a robust foundation for analyzing spatial reachability and evaluating strategies to improve it.

This methodology applies to datasets with diverse data quality and scale. It allows for adjustable spatial resolution, connection range, and sampling based on the use scenario. For example, one may choose finer resolutions to study more densely populated urban neighborhoods or coarser ones to analyze sparse areas, such as large highways. Similarly, the sampling rate opens the possibility of modeling interactions that are volatile and change during smaller time intervals or systems with more continuous mobility, whose analysis would require fewer computational resources. The final output is a "temporal" graph, where each node represents an occupied tile and each edge represents connectivity between regions at a given time. This data structure is the basis for computing time-respecting paths, constructing reachability graphs, and performing spatiotemporal analysis.

4. Evaluation

Our evaluation has two objectives: first, to analyze spatiotemporal patterns revealed by the geotemporal graph representation; second, to assess the performance of our source region selection algorithm. We describe the datasets, experimental setup, and evaluation metrics used to validate our approach.

4.1. Datasets

Real vehicular trace. The first dataset we consider is πNEUMA, an urban vehicular dataset collected to support high-resolution research on traffic dynamics and congestion in large-scale, real-world settings [16]. In contrast to traditional traffic data, typically extracted from GPS devices, the novel methods offered by πNEUMA achieve unprecedented spatial and temporal granularity. The goal was to create the most comprehensive urban dataset and to overcome the limitations of earlier approaches, which are particularly significant in dense urban environments, such as areas with poor coverage and low accuracy. The πNEUMA experiment was conducted over the central business district of Athens, covering a congested area of approximately 1.3 km^2. This region encompasses more than 100 kilometers of urban road network, 100 busy intersections, multiple public transit stops, and dense vehicular traffic characteristic of metropolitan cores. A swarm of 10 drones hovered over the target region for numerous days, continuously recording traffic movement. The results extracted from the observations comprise approximately half a million vehicle trajectories that capture flow and interactions between vehicles with outstanding accuracy. πNEUMA provides a complete time-stamped vehicular trace that offers full visibility of vehicles at any given moment. Such characteristics make this data well-suited for investigating vehicular networks, congestion, and traffic phenomena and for exploring mobile network abstractions, reachability, and message dissemination strategies. To the best of our knowledge, it remains one of the most complete datasets of its kind.

Synthetic vehicular trace. To introduce diversity in our experiments, we also consider a synthetic vehicular trace dataset generated by the SUMO traffic simulator [15]. SUMO provides a suite of tools for generating vehicular networks using urban maps. The *OSMWebWizard* automates the process of network drawing by extracting road segments and connections from OpenStreetMaps' API. This *Wizard* also automates traffic generation based on a specific probability distribution. For each mode of transport, the Through Traffic Factor and Count parameters influence network vehicle insertion. Additional parameters enable manual adjustments to trip and route creation.

4.2. Evaluation Setup

For the πNEUMA dataset, we evaluated all available regions of Athens. We constrain our analysis to a single day (October 24, 2018) and the half-hour interval from 8:00 am to 8:30 am. This period captures a typical morning traffic pattern that offers a good balance between node density and movement diversity. Due to the vehicle's low displacement during this period, we sampled the data at a temporal resolution of one second. The mean speed supports the supposition that connectivity events do not change over short intervals.

For the SUMO dataset, we used *OSMWebWizard* to extract the vehicular network from the 15th district of Paris. This area, with an approximate size of 4.5 km^2, falls within a bounding box defined by coordinates (48.834581790851956, 2.2780808448789998) and

(48.8479556182268, 2.31927957534782). To mimic real traffic conditions as closely as possible, we tuned the traffic generation parameters extracting data from the πNEUMA dataset. For each vehicle type, we computed the maximum speed, maximum acceleration, maximum deceleration, insertion ratio, and fringe ratio. The insertion ratio corresponds to the percentage of vehicles of that type, and the fringe ratio corresponds to the percentage of vehicles that start or end their path at the region's borders. The final parameters used in route and trip generation with the *randomTrips* tool appear in Table 1. The movement statistics apply directly, while the ratios composed the insertion rate and fringe factor. Due to the high-density scenario in the πNEUMA dataset, which resulted in unrealistic traffic jams, we scaled the traffic by a factor of 0.5 over the 2,000-second simulation period.

Table 1: *randomTrips* parameters for trips and routes generation for constructing the SUMO dataset.

	Vehicle Type			
	Bus	Motorcycle	Passenger	Truck
fringe-factor	6	6	6	6
insertion-rate	700	11,200	21,000	1,820
maxSpeed	9.00	13.93	11.23	8.33
accel	1.34	2.31	1.83	1.51
decel	1.38	2.09	1.81	1.50
sigma	0.5			
departLane	best			
departPos	last			
departSpeed	avg			
max-distance	2,400			

To compare the structures of both datasets, we calculated their vehicle density, maximum traveled distance, and average vehicle lifetime. The results in Table 2 reveal that the two datasets represent different situations. As the SUMO dataset covers a larger area, it presents a situation where vehicles travel longer distances but proportionally take longer to do so when compared to the πNEUMA Dataset, indicating a different type of congestion where vehicles drive slower or spend more time stopped.

Table 2: Metrics comparison between the two datasets.

	Dataset	
	πNEUMA	SUMO
Density (veh/km^2)	7,917	537
Avg. traveled distance (m)	494	1,729
Avg. lifetime (s)	125	573

The communication range is 50 m. This decision relies on prior work in high-speed device-to-device connections, which utilized empirical measurements to determine the link's goodput as a function of distance. We established a goodput lower threshold of 10 mega bytes MBytes/s, giving us the connection limit at 50 m [18].

The spatial resolution used to index each tile by H3 was resolution 11. At this level, the average edge length of the hexagonal tiles is approximately 28.87 meters, which ensures that the distance between two adjacent tiles matches the connection range. We selected this resolution so that tiling abstracts the network by representing more than a single node, but not too large, which would mean covering more non-road areas than roads. Given the average tile-to-tile center distance at resolution 11, this communication range results in a hop threshold $\theta = 1$, meaning that two tiles are connected at a given time if the minimum number of adjacent-tile transitions between them is one (i.e., they are adjacent).

The communication delay tolerance δ characterizes the urgency of the message dissipation. We evaluated different latency constraints to effectively suit diverse use cases, modeling each one at a specific priority level. In vehicular delay tolerant networks, non real time applications such as road safety, traffic flow optimization, marketing advertising, and environmental sensing tolerate long delays [19]. Therefore, the chosen values are 5, 10, 15, 20, 30, 60, 90, 120, and 150 seconds, which represent a good gradation and a diversity of use cases.

4.3. Performance Assessment

We measure the effectiveness of a given source set $\mathbb{S}$ for a certain delay tolerance δ by calculating a new metric called *Space-aggregated Reachability*, here denoted as $\mathcal{SR}_\delta(\mathbb{S}, t)$. At each time step t, we use the previously defined reachability graph for one delay δ and count how many tiles the source set can reach. We account for each tile once, regardless of how many source tiles can reach it. Formally, the space-aggregated reachability of a source set, for a given delay and time, is defined as:

$$\mathcal{SR}_\delta(\mathbb{S}, t) = \frac{|\{v \in V_t \mid \exists u \in \mathbb{S}, (u, v) \in \mathcal{R}_\delta(t)\}|}{|\bigcup_{t \in (t, t+\delta)} O_t|}. \quad (5)$$

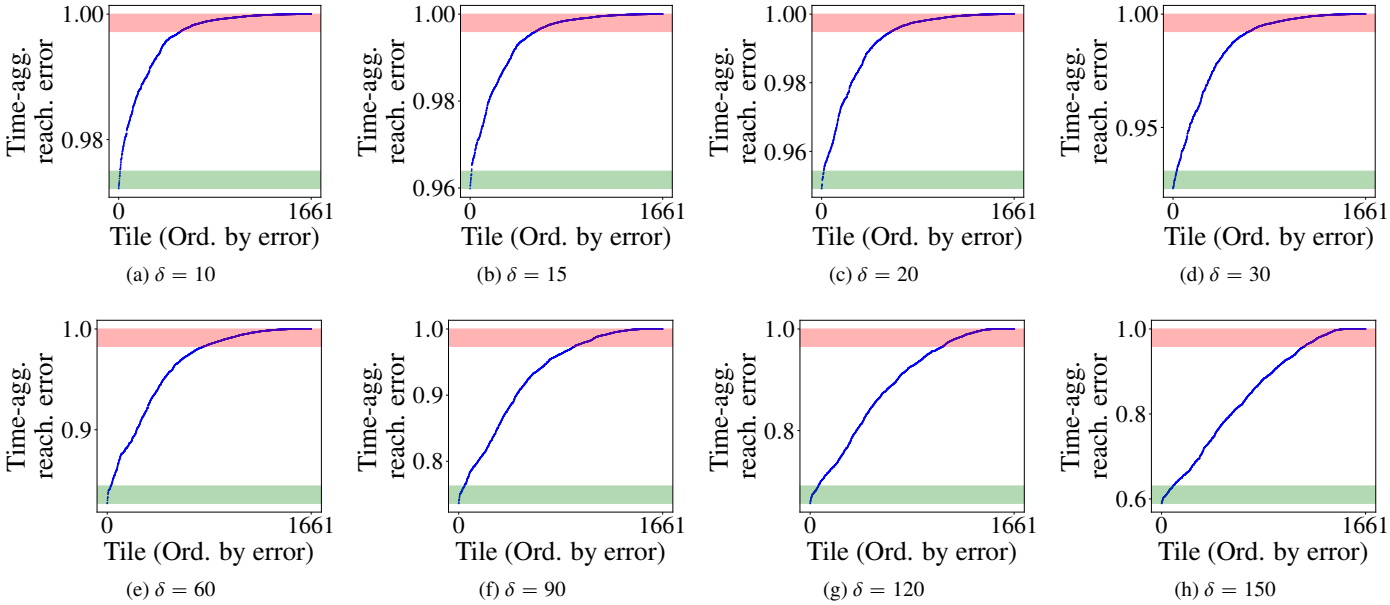
For each time step, this function outputs the proportion of all occupied tiles in the period $(t, t + \delta)$ that are reached by a message that starts from any selected source at t and has δ seconds to spread.

5. Computing Tile Reachability

In this section, we investigate the reachability from source tiles. We identify a small subset of tiles that consistently outperforms others in temporal reachability, and subsequently present the structural characteristics of this subset. By evaluating reachability-based metrics, such as time-aggregated reachability error and topological characteristics, we aim to identify spatial patterns related to connectivity behavior over time. These insights form the foundation for extracting the spatial knowledge necessary to develop efficient source selection algorithms that enhance reachability and improve overall performance.

5.1. Numerical Results

Figures 4 and 5 present the plots of time-aggregated reachability error for eight distinct values of δ across the entire observed time window for SUMO and πNEUMA, respectively. The red and green regions represent two tile subsets: those with



(a) $\delta = 10$ (b) $\delta = 15$ (c) $\delta = 20$ (d) $\delta = 30$

(e) $\delta = 60$ (f) $\delta = 90$ (g) $\delta = 120$ (h) $\delta = 150$

Figure 4: SUMO dataset. Time-aggregated reachability error for each tile, ordered by error, using five different δ (latency constraints).

Table 3: Descriptive statistics for candidate and ineffective sets in function of δ (SUMO dataset).

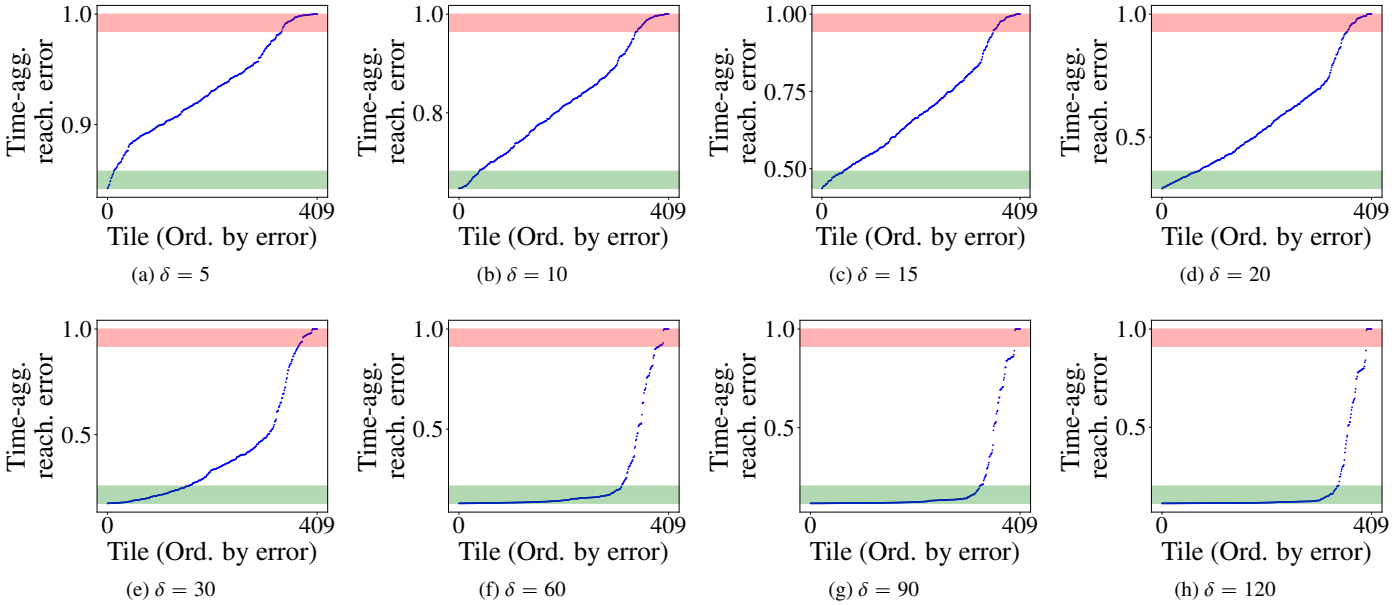
δ	**Set**	**Latitude (mean ± std)**	**Longitude (mean ± std)**	**Size**	**Score (mean ± std)**
10	*Candidate*	48.8406 ± 0.0021	2.3016 ± 0.0040	13	0.9735 ± 0.0008
	Ineffective	48.8414 ± 0.0047	2.2978 ± 0.0126	1,128	0.9994 ± 0.0007
15	*Candidate*	48.8399 ± 0.0008	2.3000 ± 0.0006	13	0.9618 ± 0.0011
	Ineffective	48.8414 ± 0.0047	2.2977 ± 0.0127	1,097	0.9991 ± 0.0010
20	*Candidate*	48.8406 ± 0.0015	2.2992 ± 0.0020	19	0.9518 ± 0.0015
	Ineffective	48.8414 ± 0.0047	2.2976 ± 0.0128	1062	0.9988 ± 0.0013
30	*Candidate*	48.8415 ± 0.0017	2.2990 ± 0.0025	32	0.9270 ± 0.0022
	Ineffective	48.8413 ± 0.0048	2.2975 ± 0.0129	1,007	0.9980 ± 0.0021
60	*Candidate*	48.8419 ± 0.0017	2.2999 ± 0.0019	31	0.8388 ± 0.0047
	Ineffective	48.8412 ± 0.0049	2.2975 ± 0.0133	875	0.9951 ± 0.0052
90	*Candidate*	48.8419 ± 0.0015	2.2997 ± 0.0029	48	0.7534 ± 0.0067
	Ineffective	48.8413 ± 0.0051	2.2971 ± 0.0139	716	0.9928 ± 0.0080
120	*Candidate*	48.8421 ± 0.0016	2.2993 ± 0.0035	67	0.6772 ± 0.0083
	Ineffective	48.8410 ± 0.0053	2.2967 ± 0.0140	579	0.9915 ± 0.0097
150	*Candidate*	48.8419 ± 0.0016	2.2994 ± 0.0041	94	0.6129 ± 0.0106
	Ineffective	48.8411 ± 0.0054	2.2969 ± 0.0142	500	0.9893 ± 0.0127

the highest 10% reachability error, referred to here as the *ineffective set*; and those with the lowest 10%, the *candidate set*. Tables 4 and 3 display corresponding metrics for both sets announced above. To explore the geographic disposition of tiles, we compared the two-dimensional mean and standard deviation of tile positions between each set. Additionally, the table includes the size of each set, along with the mean and standard deviation of the associated scores.

5.2. SUMO Dataset Analysis

Figure 4 illustrates that, as anticipated, less time-sensitive messages reach a greater number of tiles due to the extended duration available for dissemination, reducing the overall error. Despite this trend, the plots also reveal that all tiles exhibit poor performance for all δ values, and even under relaxed latency constraints, the error remains above 0.65. This indicates a low reachability condition across the synthetic dataset. We also observe persistent disparities, where a set of tiles remains ineffective through all latency scenarios. These observations highlight the characteristics of the synthetic dataset, revealing a disconnected vehicular network in a less congested situation, leading to limitations in message spreading.

To better understand this disparity, we analyzed the characteristics of two tile subsets: those with the highest 10% reachability error, referred to here as the *ineffective set*, and those with the lowest 10%, the *candidate set*. The results in Table 3 underline that *candidate tiles* are closer to each other when compared with the *ineffective set*. Figure 6 shows that *candidate tiles* usu-



(a) $\delta = 5$ (b) $\delta = 10$ (c) $\delta = 15$ (d) $\delta = 20$

(e) $\delta = 30$ (f) $\delta = 60$ (g) $\delta = 90$ (h) $\delta = 120$

Figure 5: πNEUMA dataset. Time-aggregated reachability error for each tile, ordered by error, using five different δ (latency constraints).

ally represent intersection areas, and ineffective cells remain in the margins of the region.

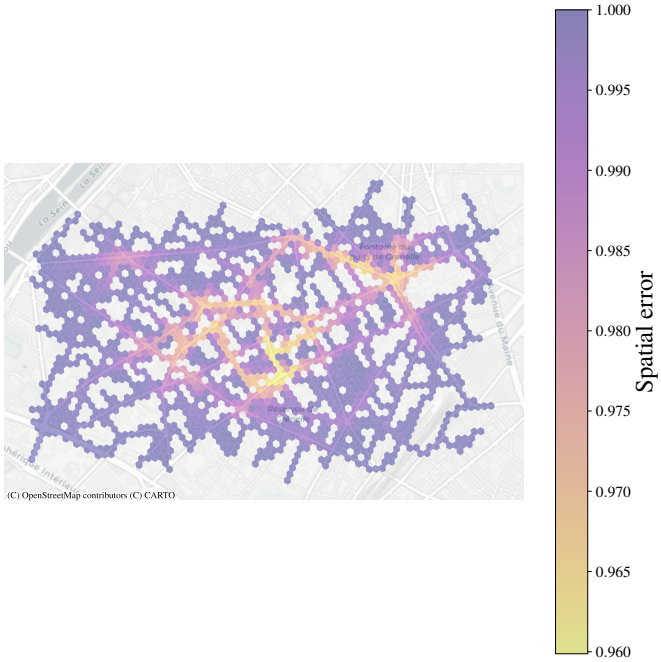


Figure 6: SUMO dataset. Time-aggregated reachability error map for $\delta = 15$.

A notable phenomenon is the impact of the delay constraint on the *ineffective set*. The size of this set not only decreases for higher δ, but also the standard deviation of the error augments, as shown in Table 3. This suggests that with less urgent messages, the structure of this set changes and fewer tiles are ineffective, as displayed by the diminution of the plateaus in Figure 4.

For smaller values of δ, the *candidate set* displays a reduced reachability when compared against the *ineffective set* (the decrease of mean time-aggregated reachability error is less than 0.1). For latencies greater than one minute, the mean error becomes more pronounced, and the set also becomes larger. These observations suggest that scenarios with delays in the range of minutes will result in a greater increase in reachability. Nevertheless, as we do not know how fair those distributions are or how well they complement each other in terms of union and intersection, the selection of a good source set does not rely solely on choosing the best tiles among the *candidate set* and requires a more sophisticated algorithm.

5.3. πNEUMA Dataset Analysis

For the real-world dataset, Figure 5 shows that less urgent messages also reach more tiles for the same reason. Additionally, the disparities among tiles persist, with some regions still presenting extremely low reachability, even with relaxed latency constraints. In contrast, tiles achieve low error rates with smaller δ; this difference highlights that, as a consequence of the inherent network disconnections, the higher density scenario influences message reachability.

To better understand this new condition, we proceed to analyze the two characteristic tile subsets. The results in Table 4 show that *candidate tiles* repeat the pattern of being situated closer to each other when compared with the *ineffective set*, especially for low latencies. Figure 7 reinforces that *candidate tiles* usually represent intersection areas, here in the middle of the region, and ineffective cells remain in the margins of the region.

In the πNEUMA Dataset, both sets respond differently to the delay constraint. With higher latencies, the size of the *ineffective set* decreases, but importantly, a few tiles remain with an error of 1.0. Furthermore, the size of the *candidate set* not

Table 4: Descriptive statistics for candidate and ineffective sets in function of δ (πNEUMA Dataset).

Delta	Set	Latitude (mean ± std)	Longitude (mean ± std)	Size	Score (mean ± std)
5	*Candidate*	37.9865 ± 0.0008	23.7289 ± 0.0005	13	0.8500 ± 0.0048
	Ineffective	37.9839 ± 0.0047	23.7317 ± 0.0040	70	0.9965 ± 0.0037
10	*Candidate*	37.9860 ± 0.0014	23.7296 ± 0.0009	39	0.6615 ± 0.0103
	Ineffective	37.9840 ± 0.0048	23.7316 ± 0.0041	65	0.9892 ± 0.0109
15	*Candidate*	37.9851 ± 0.0015	23.7298 ± 0.0010	45	0.4693 ± 0.0162
	Ineffective	37.9842 ± 0.0050	23.7318 ± 0.0041	56	0.9831 ± 0.0161
20	*Candidate*	37.9845 ± 0.0016	23.7300 ± 0.0016	71	0.3318 ± 0.0207
	Ineffective	37.9843 ± 0.0051	23.7314 ± 0.0040	48	0.9784 ± 0.0196
30	*Candidate*	37.9847 ± 0.0024	23.7301 ± 0.0025	155	0.2075 ± 0.0243
	Ineffective	37.9837 ± 0.0052	23.7317 ± 0.0039	37	0.9721 ± 0.0239
60	*Candidate*	37.9846 ± 0.0044	23.7314 ± 0.0033	318	0.1494 ± 0.0182
	Ineffective	37.9841 ± 0.0054	23.7314 ± 0.0038	19	0.9667 ± 0.0383
90	*Candidate*	37.9846 ± 0.0044	23.7314 ± 0.0032	330	0.1301 ± 0.0136
	Ineffective	37.9837 ± 0.0051	23.7310 ± 0.0034	11	0.9993 ± 0.0019
120	*Candidate*	37.9846 ± 0.0044	23.7314 ± 0.0033	343	0.1255 ± 0.0141
	Ineffective	37.9837 ± 0.0051	23.7310 ± 0.0034	11	0.9993 ± 0.0019

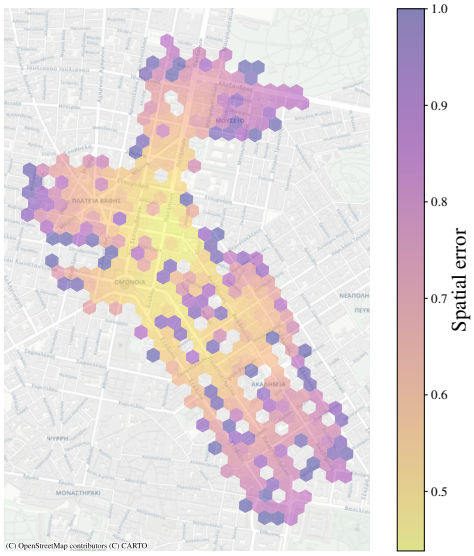


Figure 7: πNEUMA dataset. Time-aggregated reachability error map for $\delta = 15$.

only increases, but the error's standard deviation is significantly smaller, represented by the "plateau" regions in Figure 5. This suggests that with less urgent messages, a higher number of tiles is well-suited for spreading messages.

6. Source Set Analysis

This section presents the results of the proposed source set selection algorithm. First, we evaluate its performance in space-aggregated reachability to state its efficiency. Secondly, we proceed to analyze the reachability of the source set, extracting network insights. Specifically, we examine how effectively different numbers of source tiles disseminate information within specified latency and how this affects stability, redundancy, and scalability.

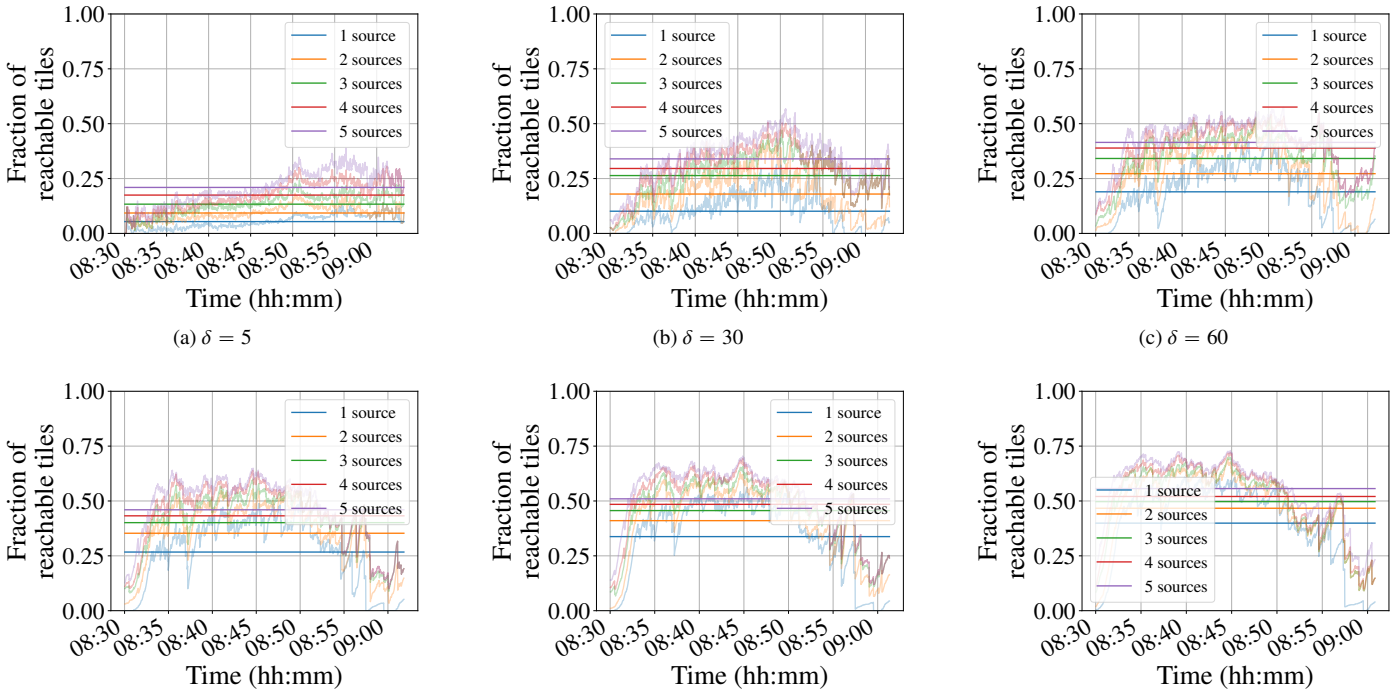
6.1. Numerical Results

Figures 8 and 9 depict the space-aggregated reachability error for the SUMO and πNEUMA datasets, respectively. For the sake of readability, the average value is shown for each source set size, while its true value is maintained with low opacity. For the same reason, we display only five representative source set sizes. Figures 10 and 9 display the space-aggregated reachability from the point of view of the latency tolerance. For each combination of δ and source set size, we quantify the space-aggregated reachability performance by calculating the average over the target period, with standard deviations represented by error bars. Each curve represents the performance of a set size related to a delay tolerance.

6.2. SUMO

Figure 10 demonstrates that the heuristic selection strategy is effective for the SUMO dataset. Notably, for all explored latency constraints, selecting additional sources significantly enhances reachability. When increasing from one to two sources is 72.6%, 77.3%, 32.1%, and 17% for δ values of 5, 30, 90, and 150 seconds, respectively. A further increment from 2 to 3 sources yields minor yet notable gains of 43.28%, 46.9%, 13.7%, and 6.5%, respectively. When expanding to 4 sources, the improvements continue to benefit especially urgent communication scenarios, with reachability increasing by 31%, 12.2%, 7.7%, and 4.7% for the same δ.

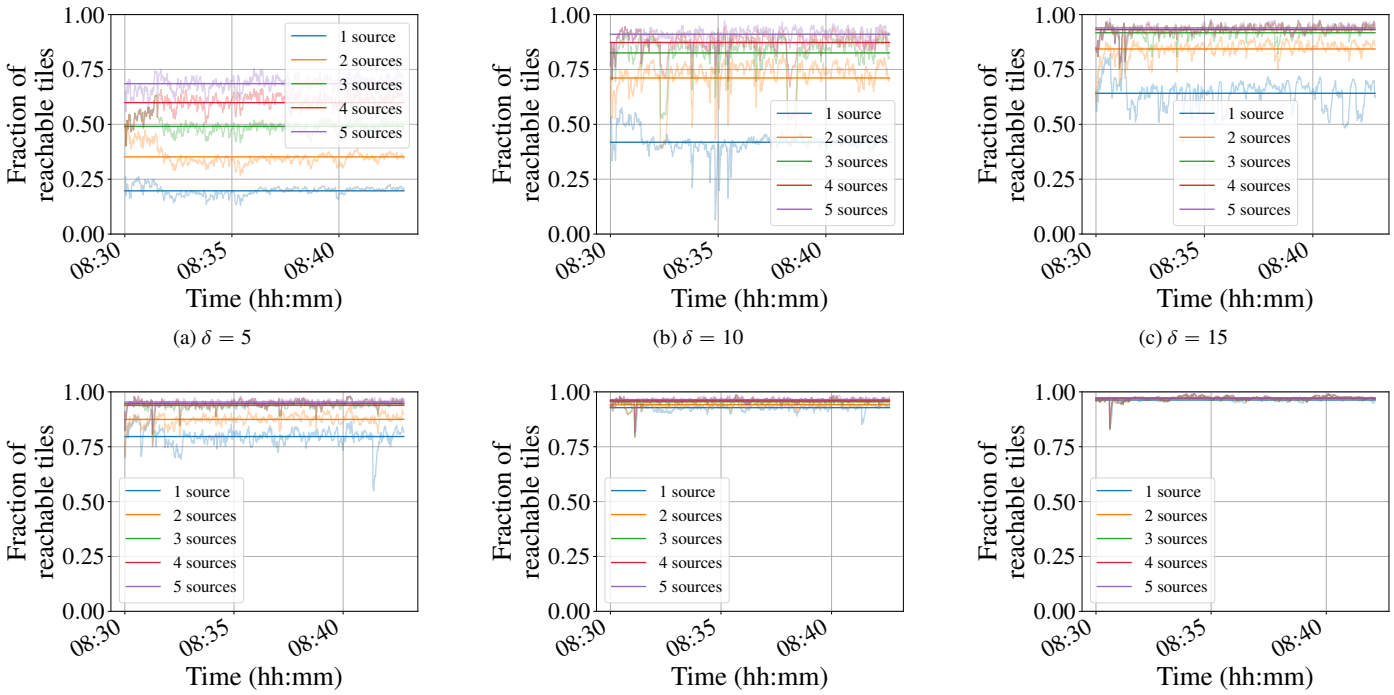
When considering the scenario with a single source tile, we observe improvement in reachability as δ increases. For δ up to 20 seconds, figure 10 shows that reachability increases with instability. As we observed in section VI, the time-aggregated reachability error decreases approximately 1% for these latency constraints. However, increasing latency from 5 to 30 s yields a 87.04% gain in coverage, followed by 87.4% from 30 s to 60 s, 41% from 60 s to 90 s, 26.3% from 90 s to 120 s, and 18.2% from 120 s to 150 s. This result aligns with the intuitive assumption that messages that have more time before expiration



(a) $\delta = 5$ (b) $\delta = 30$ (c) $\delta = 60$

(d) $\delta = 90$ (e) $\delta = 120$ (f) $\delta = 150$

Figure 8: SUMO dataset. Space-Aggregated Reachability for a source set of size up to 5 and $\delta = \{5, 30, 60, 90, 150, 180\}$ seconds.



(a) $\delta = 5$ (b) $\delta = 10$ (c) $\delta = 15$

(d) $\delta = 20$ (e) $\delta = 30$ (f) $\delta = 60$

Figure 9: πNEUMA dataset. Space-Aggregated Reachability for a source set of size up to five and $\delta = \{5, 10, 15, 20, 30, 60\}$ seconds.

can spread to more tiles. Indeed, as Figure 10 shows, the average fraction of reachable cells greatly increases with δ, which reflects the growth in reachability.

For a source set solution to serve as a message dissipation tile, it is essential to analyze the stability of its reachability. For instance, a source set that has peaks of 100% spatial reacha-

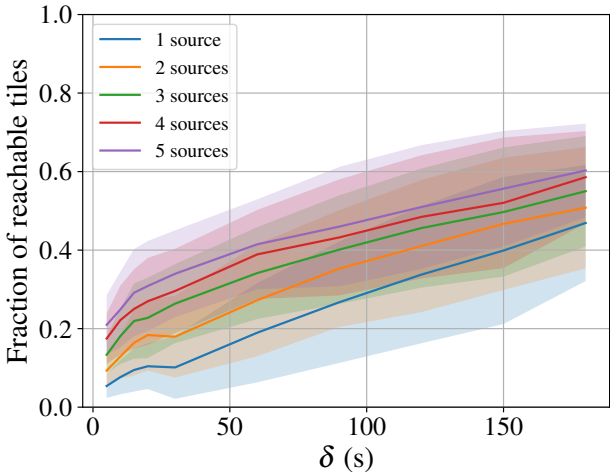


Figure 10: SUMO dataset. Average of Space-Aggregated Reachability for source set of size up to 5 and $\delta = \{5, 10, 15, 20, 30, 60\}$ seconds. Standard deviation appears as the area along the curve.

bility preceded by valleys of 0% may present an error that is below average. Still, in practical terms, it poses a problem related to timing. Relying on a service with such variation may cause important messages to reach only a small portion of the tiles in a "bad luck" scenario. The synthetic scenario exhibits an unstable behavior. As latency tolerance increases, reachability variance across time continues to increase, albeit with greater fluctuation. For example, Figure 8 illustrates that for all source set sizes, reachability fluctuates strongly when increasing the delay constraint, but becomes noticeably higher. When δ is greater than 90 seconds, the standard deviation starts to increase with lower rates until it reaches a stabilization point at 180 seconds, after which the instability begins to decrease for all sources. When going from 150 s to 180 s, the fluctuation drops 4.7%, 1.3%, 9.4%, and 3.9% for 1, 2, 4, and 5 sources, respectively. In addition, using more still causes instability growth but with smaller ratios. These results show that scenario density or even the characteristics of the flow, in this case, created by the uneven Parisian road structure, impact the stability of the network.

Crucially, increasing the number of sources allows messages to reach more tiles *within the same latency constraint*. For instance, suppose that a vehicle in a particular tile disseminates a message with a delay tolerance of 40 seconds. If the average time-aggregated reachability for $\delta = 10$s of that tile is 80%. In that case, this means that the message will arrive in almost all tiles with a maximum delay of 10 seconds, even though the tolerance is larger. This situation is evident in Figure 10, where larger source sets achieve comparable levels of reachability much earlier in time. In this case, a message with a tolerance of 150 s reaches 40% of the reachable tiles after the entire time window, whereas when spread with five sources, it reaches the same 40% in 50 seconds. Therefore, strategically selecting multiple source cells not only improves stability and overall reachability but also ensures that critical messages are delivered *faster*, which can be decisive in time-sensitive applications such as real-time alerts or emergency communications.

6.3. πNEUMA

Figure 4 shows that the heuristic choice remains appropriate for the pneuma dataset. We observed that, especially in situations with low-latency constraints, choosing the next source improves the set reachability. The growth in average space-aggregated reachability when considering two sources instead of one, for δ of 5, 10, 15, and 20, is respectively 78.5%, 70%, 31.5%, and 9.8%. When going from 2 sources to 3, the gains are diminished, but still present. For δ of 5, 10, 15, and 20, the increase is respectively 39.5%, 16.0%, 8.8%, and 7.29%. The change to four sources impacts more urgent situations – for δs of 5, 10, 15, and 20, space-aggregated reachability increases by 22.1%, 5.7%, 1.5%, and 0.8%, respectively. In scenarios with a latency constraint larger than 30 seconds, as performance converges for all set sizes, choosing more tiles to act as sources has a negligible impact on performance.

In the single-source scenario, we also observe a substantial improvement when the latency constraint increases, which also confirms the intuitive assumption. A 112.4% improvement when going from a message of 5 to 10 seconds tolerance, 53.3% from 10 s to 15 s, 24.2% from 15 s to 20 s, 16.5% from 20 s to 30 s, and 3.8% from 30 s to 60 s. The real-world data presents a more accentuated gain in reachability compared to the synthetic data, and reaches a saturation point quickly, where almost all reachable tiles receive the message. This vehicular network presents a high level of congestion and vehicle density, which may facilitate easier dissemination. Furthermore, the Athenian street network has a more grid-like structure, which leads to more stable traffic flow and also improves the reachability.

Concerning instability, the real-world case behaves differently. We observe an apparent decrease as the latency constraint and the source set size increase. For smaller δ, such as 5 to 30 seconds, spatial reachability shows significant fluctuation, reflecting frequent moments of disconnection. However, in those values of δ, adding more source tiles contributed to instability mitigation, as shown in Figure 5b, by reducing reachability valleys. Overall, while using two sources led to a mean increase of 3.9% in standard deviation, using three sources instead of two led to a mean decrease of 10.6%, from three to four sources to 6.5%, and from five sources to 6.5%. This increase in stability emphasizes the importance of using multiple source tiles, especially in tighter latency tolerances, due to the inherent fragmented nature of the network.

The delay reduction effect persists for the πNEUMA dataset. Figure 11 reveals that, while augmenting source set size may not increase spatial reachability at higher δ, it results in reaching more tiles with reduced latencies. For instance, a message spread from five sources takes 10 seconds to reach approximately 90% of occupied tiles, while it takes 30 seconds to achieve the same percentage when spread by one source.

7. Related Work

The problem of maximizing message reachability in mobile networks has garnered substantial attention in recent years, particularly when time-constrained dissemination is crucial. Existing research has primarily focused on using node-level features

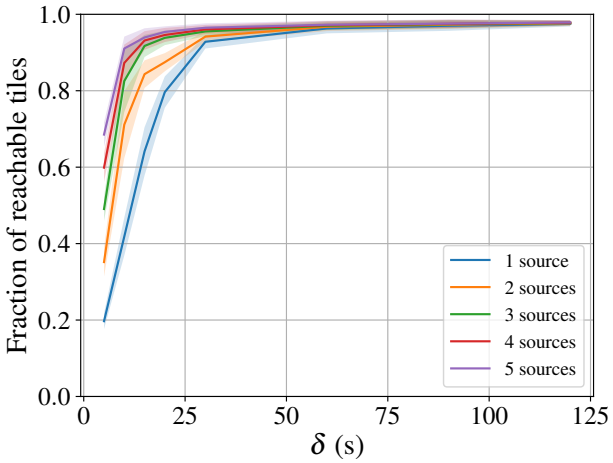


Figure 11: πNEUMA dataset. Average of Space-Aggregated Reachability for source set of size up to 5 and $\delta = \{5, 10, 15, 20, 30, 60\}$ seconds. We represent standard deviation as the area along the curve.

to identify key spreaders throughout the network. Epidemic Routing [20] and PRoPHET [21] protocols enhance message delivery ratios by utilizing historical encounters and probabilistic forwarding models. These methods frequently encounter scalability issues due to redundant transmissions, as they do not account for spatial constraints [22].

Some studies have investigated the identification of influential nodes using topological metrics, such as betweenness and eigenvector centrality, which operate on limited network information [23]. The methods demonstrate effectiveness in specific cases, yet they maintain a focus on nodes rather than fully exploiting spatial heterogeneity and regional mobility patterns.

A complementary line of work has explored the use of AI-driven intent-based techniques to optimize message dissemination policies under uncertainty. For example, Ahmad et al. propose a framework that adapts forwarding decisions based on learned safety and flow intents [24]. While such methods offer promise in optimizing V2X communications, they still operate under a node-centric model and require extensive training data to achieve optimal results. In contrast, our geotemporal approach operates independently of message content and seeks to generalize dissemination strategies by leveraging structural spatiotemporal regularities found in human and vehicular mobility.

The canonical environment for studying message dissemination in sparse and mobile scenarios is Delay Tolerant Networks (DTNs) [25]. The routing strategies of DTNs utilize store-carry-forward mechanisms to connect areas where spatiotemporal gaps exist. A wide range of heuristics and learning-based methods aim at optimizing routing under intermittent connectivity. Some methods adopt prediction models of node movement [26], while others, such as MaxProp [27] and Spray-and-Wait [28], aim to balance delivery probability with overhead.

The use of individual trajectories instead of collective spatial properties restricts the applicability of these techniques. Furthermore, many DTN approaches assume homogeneous mobility patterns or uniform spatial distributions, overlooking the influence of infrastructure constraints such as roads or buildings that channel and bias node movement [29]. Our approach differs in that we model space explicitly and search for high-impact source regions, rather than individual forwarding agents.

With the increasing availability of data, the integration of spatiotemporal analysis into network science has gained prominence. Spatiotemporal graphs are applicable in various contexts, ranging from transportation networks to epidemiological models [30, 31, 32]. For instance, recent studies have proposed time-varying graphs to represent evolving connectivity patterns in urban mobility datasets, enabling the prediction of urban passenger traffic flows with effectiveness and efficiency [33].

Some approaches propose spatial tiling or region-based abstractions for simplification, but few incorporate this structure into the message dissemination strategy itself [34, 35]. Our work builds on the insight that spatial units—specifically geographic tiles—can be evaluated as source candidates in a principled way using both their temporal activation patterns and geographic reach. By treating mobility traces as geotemporal graphs, we enable a new class of heuristics that optimize for region-based reachability rather than node-centric metrics

8. Conclusion

We developed a systematic framework for selecting source tiles that maximize vehicular network reachability, leveraging geotemporal metrics to identify a limited set of tiles with consistent high potential for message dissemination. Our findings demonstrate that geographic factors critically determine the effectiveness of tiles as message source addresses, revealing that geotemporal graph representations offer a valuable new perspective for analyzing and understanding network dynamics.

The proposed heuristic source selection algorithm achieves high reachability solutions for small set sizes while demonstrating strong scalability as networks incorporate larger numbers of sources. We observe that system stability improves and delays decrease when operators expand both the number of sources and the allowed latency period, with congested conditions showing particularly pronounced benefits from these adjustments. This behavior suggests that strategically placing sources can effectively mitigate the challenges that high-density vehicular networks typically face.

These results highlight the crucial importance of data-driven source selection methods in ensuring high reachability in multi-hop vehicular networks. The framework proves especially valuable when network designers must address high congestion levels and meet stringent latency requirements simultaneously. By combining geographic awareness with temporal network dynamics, this approach provides network operators with practical tools for optimizing message dissemination in real-world deployment scenarios. Future work could extend this methodology to dynamic source selection that adapts in real-time to changing network conditions and traffic patterns.

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